TEACHER REFERENCE • PUBLIC PAGE

Sampling: does more mean better? — Teacher reference

Prepare the lesson, inspect the reasoning and keep the evidence limits visible.

45 minutes • Junior secondary • pairs or groups of 2–4

Before you begin

Frequency, percentages and reading a bar chart. No programming required.

> An authored 45-minute sequence using an existing synthetic inquiry. The 200-person school and its activity group are invented, not a real school survey.

Synthetic starting counts: the whole school has 200 people, with 100 choosing A; the activity group has 40 people, with 36 choosing A. These fixed counts support a paper comparison and are not a newly drawn sample.

- Check that the sampling activity opens. Arrange an operator, recorder and questioner; every learner writes an individual prediction first.
- Keep the chart axis at 0% for the main comparison. The synthetic school has 100 A responses out of 200; the group has 36 out of 40.

One-class sequence

5 min **01. Predict before asking**

Will asking all 40 group members be closer to the school’ s 50% than asking 20 randomly chosen pupils? Explain before running.

> **Teacher reasoning / look for**
>
> Collect reasons before displaying results. Distinguish “more people” from “different people have a chance to be selected”.

12 min

02. Keep size, change who is asked

Use the selected seed (default 2026) to compare school-20 and club-20. Record the A count, sample size and percentage. Save both pieces of evidence in the original tool.

> **Teacher reasoning / look for**
>
> Only the sampling frame changes. A sample’ s percentage uses its own sample size as the denominator, not 200.

10 min

03. Increase size within each frame

Compare club-40 and school-40 with the earlier cases. Which pupils are still excluded by asking only the group?

> **Teacher reasoning / look for**
>
> The full group gives 36/40 = 90%, while the full school is 50%. Surveying that entire group removes within-group sampling variability but does not fix its mismatch with the school.

10 min

04. Draw again and revise

Use the next-seed control and inspect the listed 30 repeats. Revise “more people always means more accurate”. Do not pick only the nicest result.

> **Teacher reasoning / look for**
>
> The 30 repeats begin at the displayed seed. They describe these deterministic draws, not all possible surveys. Larger random samples generally reduce variability; a single draw need not improve.

8 min **05. Plan a lunch survey**

A pupil asks only the basketball team about favourite lunches. Propose a better sampling frame, note possible non-response, then answer the exit question.

> **Teacher reasoning / look for**
>
> Look for a whole-grade sampling frame and opportunities across classes/interests. An improved plan still cannot promise an exact population percentage.

Keep this evidence

- Two saved comparisons with method, size, seed and A percentage.
- A revised claim and a lunch-survey plan with one limitation.

> Use Save evidence, Download record and Print in the original inquiry. Its local record stays in this browser; this classroom page does not read, save or import it. Paper responses can be kept separately.

Exit question

If the complete activity group gives 90%, can we claim the whole school is 90%? Why?

> **Reference reasoning**
>
> No. The observed group is not the whole school. Here the invented school proportion is 50%; increasing the group sample does not include the other pupils.

Misconceptions to discuss

- A bigger sample automatically fixes selection bias.
- A lucky random draw proves the sampling method is always better.
- Changing the chart axis changes the observations.

These statements are misconceptions, not conclusions. Ask for the missing condition or evidence.

Activity source

Sequence adapted from the existing Year 7 sampling inquiry (content g7-1), including its four comparison cases, layered hints and transfer question. The 45-minute pacing is an editorial lesson plan, not a measured learning outcome.

This guide：https://skckenneth.github.io/teaching/classroom-packs/sampling-bias/teacher/
Original activity：https://skckenneth.github.io/math-studio/